

```
import pandas as pd
import matplotlib.pyplot as plt

# Read dataset
data = pd.read_csv("earthquake_1995-2023.csv")

# Convert date to year
data["date_time"] = pd.to_datetime(data["date_time"], dayfirst=True)
data["year"] = data["date_time"].dt.year

# Count earthquakes for each year
year_count = data["year"].value_counts().sort_index()

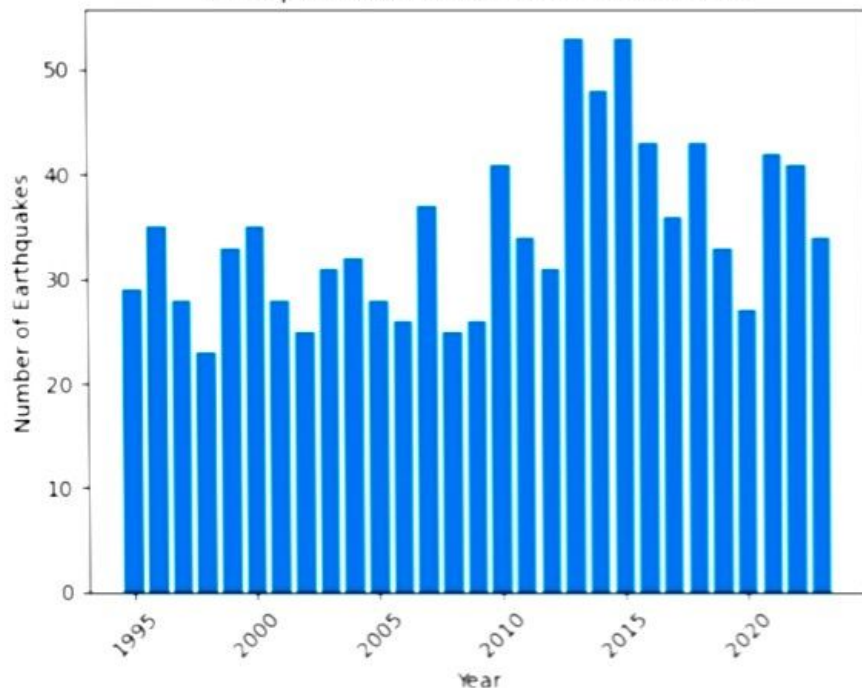
# Create bar plot
plt.bar(year_count.index, year_count.values)

plt.xlabel("Year")
plt.ylabel("Number of Earthquakes")
plt.title("Earthquakes Recorded from 1995 to 2023")

plt.xticks(rotation=45)
plt.show()
```

...

Earthquakes Recorded from 1995 to 2023



This program reads earthquake data, counts the number of earthquakes for each year from 1995–2023, and displays the results using a bar graph.



```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv("earthquake_1995-2023.csv")

data["date_time"] = pd.to_datetime(data["date_time"], dayfirst=True)
data["year"] = data["date_time"].dt.year

year_count = data["year"].value_counts().sort_index()

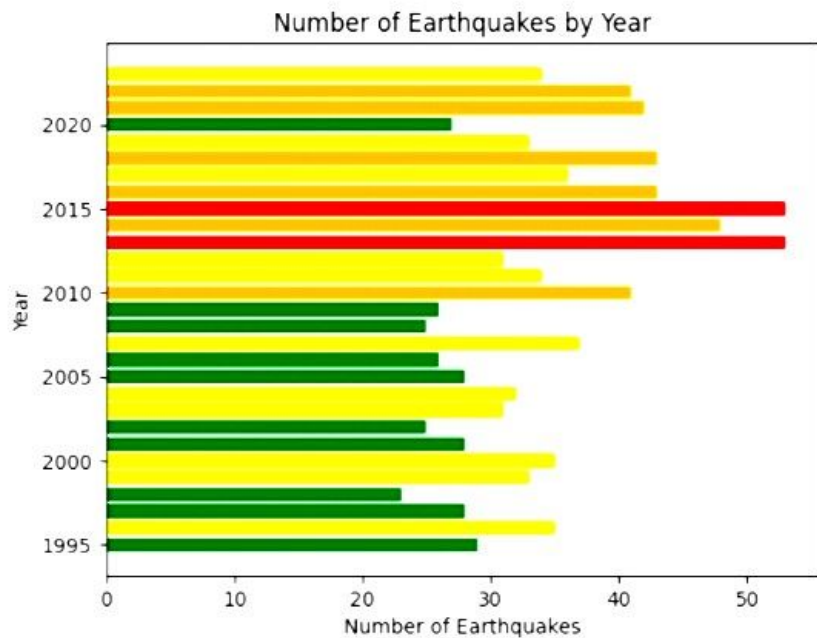
# Set colours based on earthquake level
colors = []

for count in year_count.values:
    if count < 30:
        colors.append("green")
    elif count < 40:
        colors.append("yellow")
    elif count < 50:
        colors.append("orange")
    else:
        colors.append("red")

# Horizontal bar plot
plt.barh(year_count.index, year_count.values, color=colors)

plt.xlabel("Number of Earthquakes")
plt.ylabel("Year")
plt.title("Number of Earthquakes by Year")

plt.show()
```



This program reads earthquake data, counts earthquakes for each year, and creates a horizontal bar graph where green, yellow, orange, and red represent different earthquake counts.

```
plt.figure(figsize=(8, 5))
```

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```

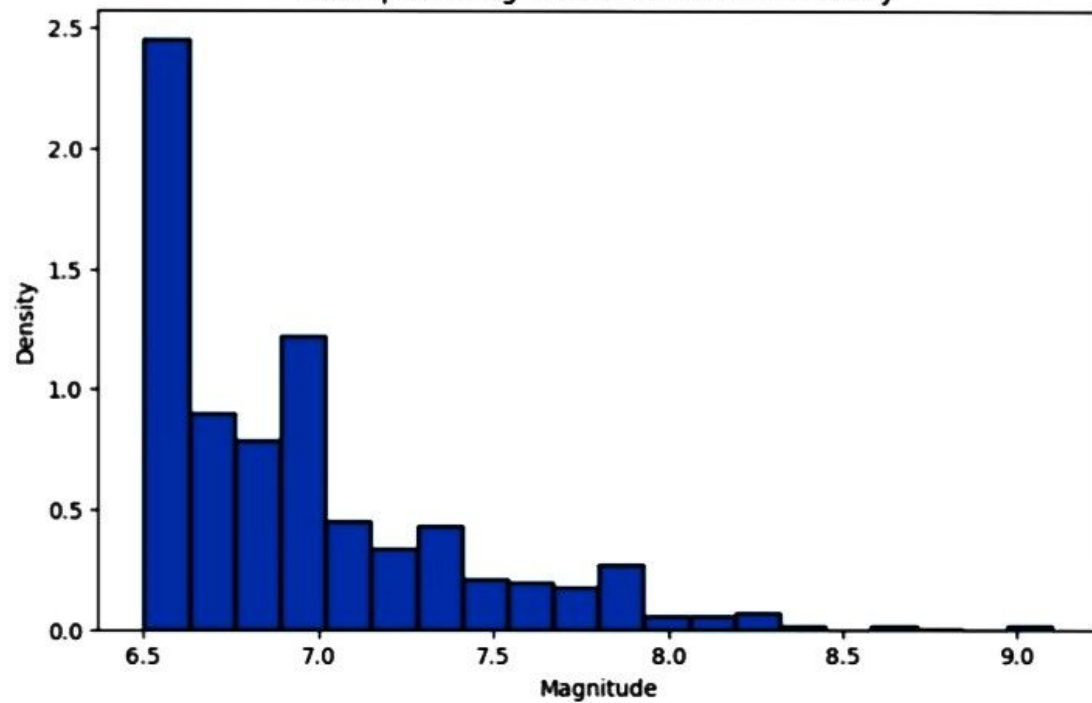
```
plt.hist(  
    magnitude,  
    bins=20,  
    density=True,  
    edgecolor="black"  
)
```

```
plt.title("Earthquake Magnitude Distribution - Density")  
plt.xlabel("Magnitude")  
plt.ylabel("Density")
```

```
plt.show()
```

...

Earthquake Magnitude Distribution - Density



This code creates a histogram showing the distribution of earthquake magnitudes using density values.



```
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("earthquake_1995-2023.csv")

# Count earthquakes with and without tsunami
tsunami_counts = df["tsunami"].value_counts()

# Create labels
labels = ["No Tsunami", "Tsunami"]

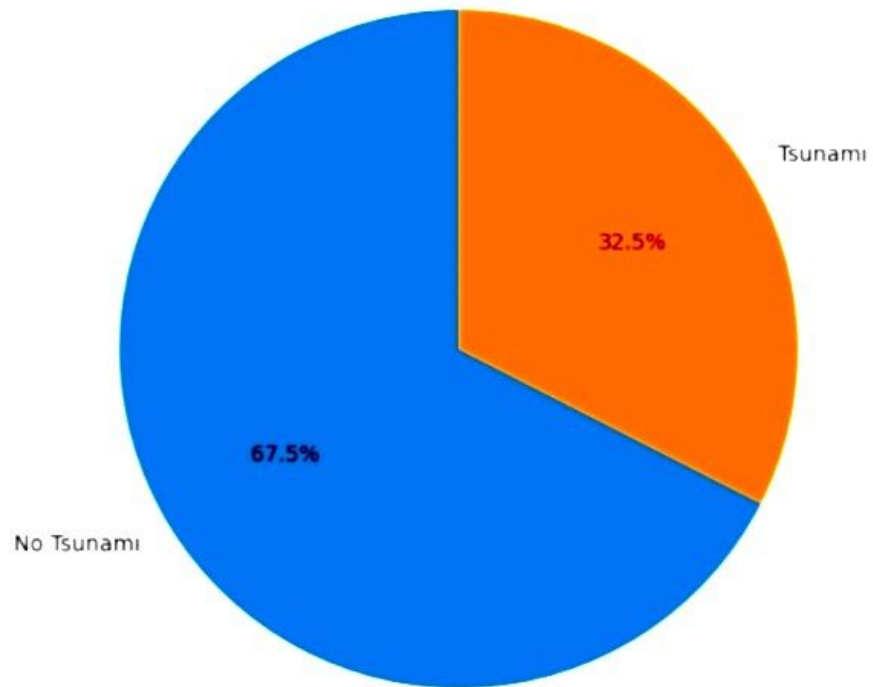
# Create pie chart
plt.figure(figsize=(7, 7))

plt.pie(
    tsunami_counts,
    labels=labels,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Earthquakes With and Without Tsunami Alerts")

plt.show()
```

Earthquakes With and Without Tsunami Alerts





This code reads the earthquake dataset and creates a pie chart showing the percentage of earthquakes with and without tsunami alerts.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("earthquake_1995-2023.csv")

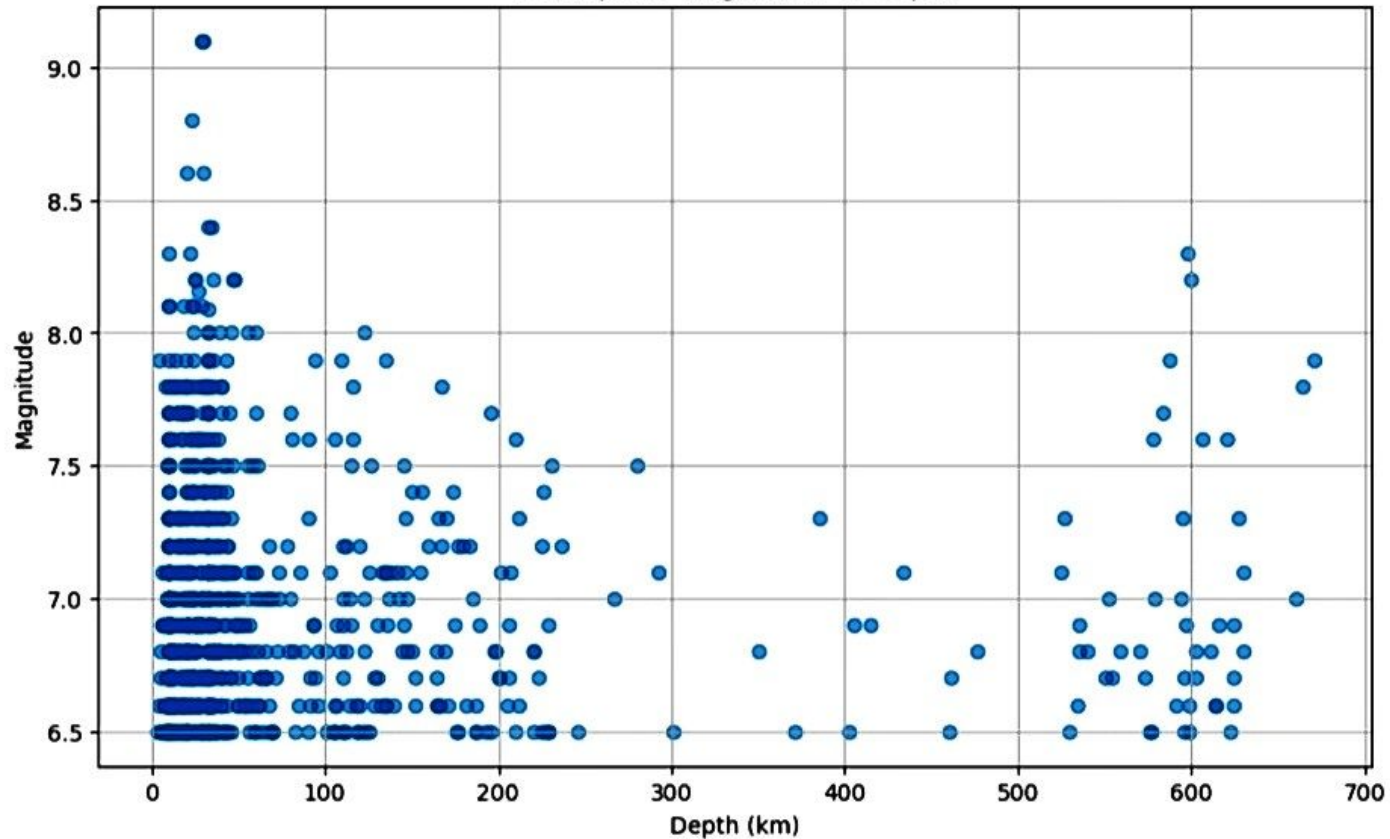
# Create scatter plot
plt.figure(figsize=(10, 6))

plt.scatter(df["depth"], df["magnitude"], alpha=0.6)

plt.xlabel("Depth (km)")
plt.ylabel("Magnitude")
plt.title("Earthquake Magnitude vs Depth")

plt.grid(True)
plt.show()
```

Earthquake Magnitude vs Depth



This code creates a scatter plot to see how earthquake magnitude is related to earthquake depth.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("earthquake_1995-2023.csv")

# Convert date_time column to datetime
df["date_time"] = pd.to_datetime(
    df["date_time"],
    dayfirst=True,
    errors="coerce"
)

# Extract year
df["year"] = df["date_time"].dt.year

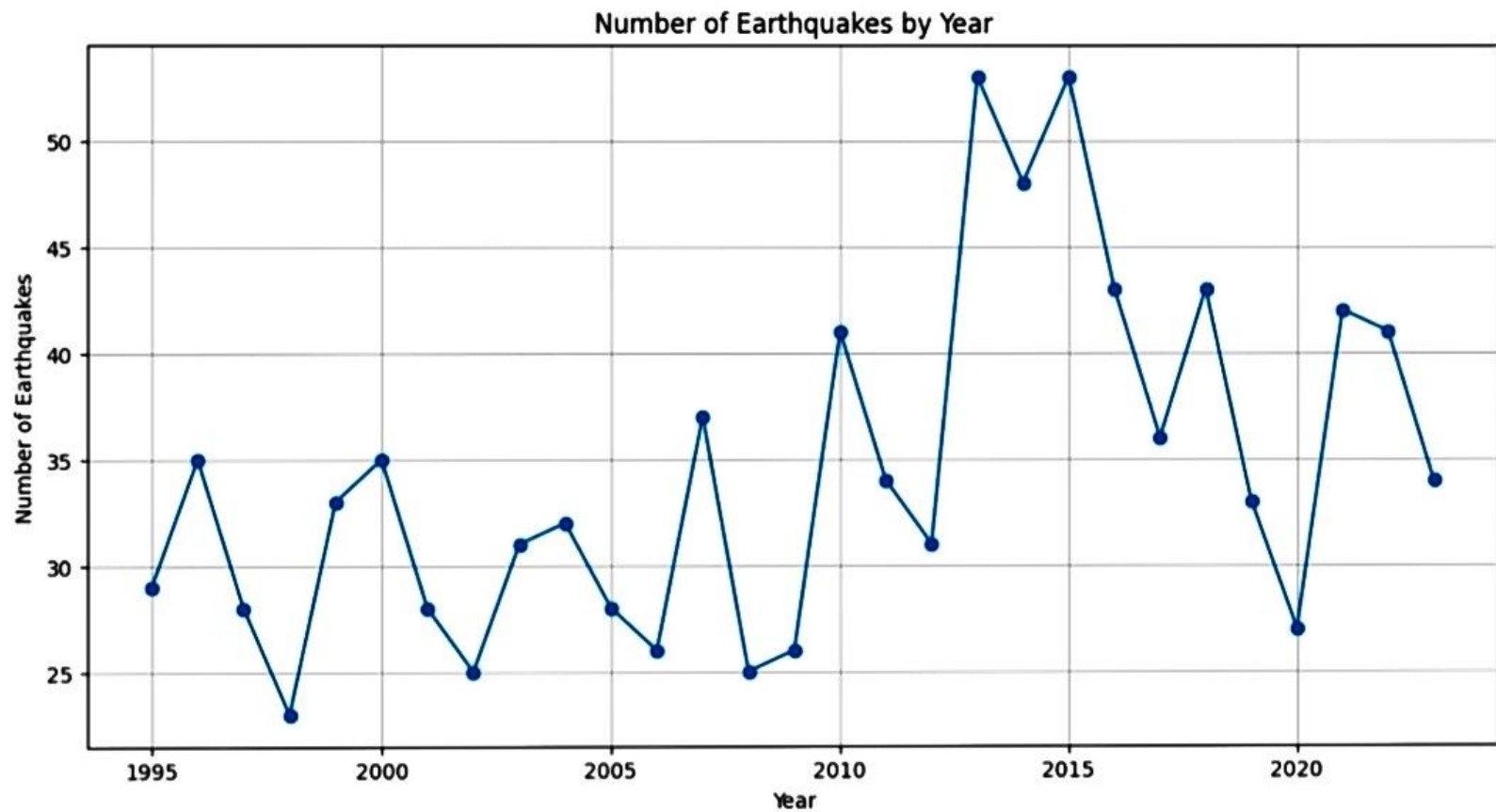
# Count earthquakes for each year
yearly_count = df["year"].value_counts().sort_index()

# Create line plot
plt.figure(figsize=(12, 6))

plt.plot(
    yearly_count.index,
    yearly_count.values,
    marker="o"
)

plt.xlabel("Year")
plt.ylabel("Number of Earthquakes")
plt.title("Number of Earthquakes by Year")
```

```
fig=plt.gcf(),  
plt.show()
```



This code counts earthquakes for each year from 1995–2023 and creates a line graph to show how the number of earthquakes changes over the years.